C-CLARKE
INTERNATIONAL INSTITUTE
OF DIGITAL SCIENCES

PROGRAM HANDBOOK

CYBER SECURITY
FAST TRACK PROGRAM

Security Operations and Identity Security

Practical, industry-oriented cybersecurity learning

6 MONTHS | PRACTICAL LABS | INDUSTRY PROJECTS



C-Clarke International Institute of Digital Sciences

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WELCOME TO THE PROGRAM

Welcome to the Cyber Security Fast Track Program. Over six months, you will develop practical capability across IT systems, networking, security operations and identity security.

This is an intensive, applied program. Progress is demonstrated through laboratories, investigations, scripts, portfolio evidence and industry-oriented projects. Use this handbook to prepare for each stage and track what you build.

YOUR LEARNING JOURNEY

BUILD	INVESTIGATE	DEMONSTRATE
Secure foundations	Events and attacks	Practical outcomes

EDUCATION FOR DIGITAL CAREERS

C-Clarke International Institute of Digital Sciences prepares learners for careers across software, technology and digital sciences. Its programs combine current technical knowledge with practical, industry-relevant learning.

Students learn with guidance from lecturers and consultants who bring professional experience into the classroom. The institute promotes innovation, ethical practice, lifelong learning and the confidence to contribute locally and globally.

VISION

To be acknowledged as a leader in digital sciences education while inspiring young leaders to achieve economic strength and independence.

MISSION

To provide innovative education that builds essential skills for professional success locally and globally.

OUR CORE VALUES

01 **Innovation**
Current industry-relevant learning

02 **Integrity**
Transparent and ethical practice

03 **Excellence**
High standards in education

04 **Empathy**
A supportive learner community



PROGRAM AT A GLANCE

An intensive six-month journey that develops foundational knowledge and applied capability across Security Operations and Identity Security.

6 MONTHS

FAST-TRACK FORMAT

Foundations, core skills and applied practice

DESIGNED FOR

- Students pursuing a cybersecurity career
- IT support, networking or systems learners
- Early-career professionals moving into SOC or IAM
- Learners who prefer practical, project-based study

01 Cyber Foundations

Systems, networks, scripting, risk and security controls.

02 Security Operations

Monitoring, investigation, detection, response and automation.

03 Identity Security

Authentication, federation, governance and privileged access.

04 Cloud and GRC

Cloud controls, risk frameworks, audit evidence and reporting.

PRACTICAL OUTCOMES

- Build and document a cyber lab
- Configure modern identity controls
- Investigate common security events
- Create an employer-ready portfolio

SECURE THE SYSTEMS PEOPLE RELY ON



REAL-WORLD SECURITY WORK
Monitor, investigate, protect and communicate

WHY IT MATTERS

Digital services depend on secure systems and trusted identities. Cybersecurity professionals help organisations reduce risk and respond effectively.

- Defend endpoints, networks, cloud services and data
- Detect and investigate suspicious activity
- Protect accounts, permissions and application access
- Turn technical evidence into clear action

INDICATIVE CAREER PATHWAYS

Opportunities depend on capability, experience and employer requirements

01 Security Operations

- SOC Analyst L1/L2
- Monitoring Analyst
- Incident Response Analyst

02 Identity and Access

- Identity Administrator
- IAM Engineer
- Federation Engineer

03 Cloud Security

- Cloud Security Engineer
- Azure Security Engineer
- AWS Security Engineer

04 Risk and Compliance

- GRC Analyst
- Risk or Compliance Analyst
- Security Audit Pathway

LEARN BY DOING

Students progress by building, configuring, investigating, documenting and presenting practical evidence - not by attendance alone.

BY COMPLETION YOU SHOULD BE ABLE TO

01 Build Infrastructure

Configure Windows, Linux, networks, virtual machines and Active Directory.

02 Apply Security Principles

Explain risk, controls, Zero Trust, NIST CSF and ISO 27001 concepts.

03 Engineer Identity

Work with Entra ID, MFA, Conditional Access, OAuth, OIDC, SAML and SCIM.

04 Operate Security

Monitor events, investigate incidents, map threats and support response.

05 Automate Tasks

Use Python, PowerShell, KQL, APIs and workflows for security operations.

06 Present Evidence

Produce diagrams, scripts, reports, runbooks and project presentations.

HOW YOU LEARN

A repeating practical learning cycle

01 **UNDERSTAND**
Guided concepts and demonstrations

02 **CONFIGURE**
Hands-on systems and identity labs

03 **INVESTIGATE**
Threat and access scenarios

04 **DOCUMENT**
Portfolio evidence and runbooks

05 **PRESENT**
Projects, reviews and feedback

BUILD • PROVE • IMPROVE

YOUR LEARNING JOURNEY

The original Months 1-9 curriculum is reorganised into one integrated six-month fast-track journey covering foundations, SOC, IAM, Cloud Security and GRC.

FOUNDATIONS AND SECURITY CORE

01 Systems and Networks

- Windows, Linux and virtual machines
- Active Directory fundamentals
- TCP/IP, DNS, DHCP and VLANs
- VPNs, firewalls and Wireshark

MILESTONE Core lab foundation

02 Security and Scripting

- CIA, risk, controls and Zero Trust
- NIST CSF and ISO 27001
- Python, PowerShell, JSON and CSV
- REST APIs, GitHub and documentation

MILESTONE Small Company Cyber Lab

03 Identity and Protocols

- Authentication and authorisation
- Identity lifecycle, RBAC and ABAC
- OAuth 2.0, OIDC, JWT and SAML
- SCIM, Keycloak and API security

MILESTONE OIDC and SAML integration

APPLIED SECURITY PRACTICE

04 Microsoft Security and Threats

- Entra ID, MFA and Conditional Access
- Defender, Sentinel and Intune
- MITRE ATT&CK and OWASP Top 10
- Phishing, malware and credential attacks

MILESTONE Identity platform or Mini SOC

05 Security Operations

- SIEM monitoring and log onboarding
- Alert triage and incident response
- Detection engineering and hunting
- Threat intelligence, SOAR and automation

MILESTONE Build and operate a Mini SOC

06 Identity, Cloud and GRC

- Hybrid identity, SSO and governance
- Application onboarding and automation
- PAM, Cloud Security and compliance
- Risk, audit evidence and reporting

MILESTONE Enterprise identity portfolio

BUILD THE TECHNICAL FOUNDATION

The first two months establish the systems, networking, security and scripting capability required for every later SOC and IAM activity.

01 SYSTEMS AND NETWORKS
FOUNDATION PHASE

M01 IT Fundamentals

- Computer architecture, operating systems and virtual machines
- Windows and Linux administration; users, services and permissions
- Active Directory fundamentals and a basic domain build

M02 Networking Fundamentals

- OSI and TCP/IP, IPv4, subnetting, TCP/UDP, ports and protocols
- DNS, DHCP, routing, switching, VLANs, VPNs, firewalls and NAT
- Addressing, troubleshooting and Wireshark packet analysis

PRACTICAL BUILD

Windows Server • Ubuntu • Active Directory • Network services

02 SECURITY AND SCRIPTING
FOUNDATION PHASE

M03 Cyber Security Fundamentals

- CIA triad, threats, vulnerabilities, risk and security controls
- Defence in depth, Zero Trust, NIST CSF and ISO 27001

M04 Scripting for Security

- Python and PowerShell fundamentals, functions and files
- JSON, CSV, REST APIs, log processing and automation

M05 Professional Digital Skills

- Git, GitHub, Visual Studio Code and technical documentation
- Ticketing, teamwork and responsible use of AI

MONTH 2 MILESTONE

Small Company Cyber Lab and portfolio demonstration

EVIDENCE Architecture diagram • AD build • Network validation • Packet analysis • Python and PowerShell scripts • GitHub portfolio

CONTROL IDENTITY. UNDERSTAND THREATS.

Months 3 and 4 connect modern identity controls with Microsoft security platforms, attack techniques and practical SOC and IAM operations.

03 IDENTITY AND PROTOCOLS
SECURITY CORE

M06 Identity Fundamentals

- Digital identity, proofing, authentication and authorisation
- Lifecycle, federation, SSO, MFA, passwordless and passkeys
- RBAC, ABAC, least privilege, Zero Trust and entitlement mapping

M07 IAM Protocols and API Security

- OAuth 2.0 roles, grants, PKCE, client credentials and consent
- OIDC, ID tokens, access tokens, JWT and token validation
- SAML 2.0 metadata, federation, claims and enterprise SSO
- SCIM, REST API security and provisioning concepts

PRACTICAL OUTPUT

Identity lifecycle, claims map, OIDC and SAML integration packs

04 MICROSOFT SECURITY AND THREATS
SECURITY CORE

M08 Microsoft Security Fundamentals

- Microsoft 365, Entra ID, users, groups and enterprise apps
- MFA, Conditional Access, Defender, Sentinel and Intune

M09 Threat and Attack Fundamentals

- MITRE ATT&CK, Cyber Kill Chain, OWASP Top 10 and malware
- Phishing, credential attacks, token theft and insider threats

M10 SOC and IAM Career Foundations

- Alert monitoring, triage, escalation and evidence collection
- IAM operations, app onboarding, tickets, troubleshooting and runbooks

MONTH 4 MILESTONE

Integrated Identity Platform and Mini SOC investigation

EVIDENCE OIDC and SAML packs • CA matrix • SOC dashboard • MITRE map • Threat investigation • Operations runbook

DETECT. INVESTIGATE. RESPOND.

Month 5 develops practical SOC capability across monitoring, incident investigation, detection engineering, threat hunting, automation and platform operations.

05 SECURITY OPERATIONS SPECIALIZATION

MONITORING • DETECTION • RESPONSE • AUTOMATION

S01 Security Monitoring and SIEM

SOC model, SIEM architecture, log sources, normalisation, correlation, alerts and dashboards.

LAB Onboard logs, build a dashboard and validate an alert

S02 Alert Triage and Investigation

Evidence preservation, incident lifecycle, timelines, root cause, escalation and containment.

LAB Investigate phishing, malware or account compromise.

S03 Detection Engineering

Detection lifecycle, analytics rules, false positives, tuning, detection-as-code and ATT&CK mapping.

LAB Create and test KQL or equivalent detections.

S04 Threat Hunting and Intelligence

IOC and IOA analysis, enrichment, hypothesis hunting, UEBA, XDR and threat intelligence.

LAB Run a structured hunt and present findings.

S05 Security Automation and SOAR

Playbooks, ticketing, enrichment, containment workflows, approvals, Python and REST APIs.

LAB Automate a phishing or suspicious sign-in workflow.

S06 SOC Platform Engineering

Connectors, parsers, health, performance, access control, content, backup and change management.

LAB Onboard a source and prepare a maintenance plan.

MONTH 5 PROJECT **BUILD AND OPERATE A MINI SOC**

Onboard logs • create detections • investigate incidents • conduct a hunt • automate one response

CONNECT. GOVERN. PROTECT.

Month 6 turns identity into a security control—linking directories, modern authentication, federation and governance through practical enterprise scenarios.

ENTERPRISE IDENTITY FLOW



I01 Hybrid Identity & Directory Services

- Active Directory and Microsoft Entra ID
- Users, groups and administrative units
- Kerberos, NTLM, LDAP/S and DNS
- Hybrid join, GPO, MFA and Conditional Access

PRACTICAL Build a hybrid identity baseline.

EVIDENCE Design + CA matrix + troubleshooting record

I02 Authentication & Federation Engineering

- OAuth 2.0, OpenID Connect and SAML 2.0
- JWT, claims mapping and token validation
- Identity brokering and federation design
- Enterprise SSO integration and diagnosis

PRACTICAL Integrate and test OIDC/SAML apps.

EVIDENCE Integration pack + claims map + test report

I03 Identity Governance & IGA Operations

- Joiner–Mover–Leaver lifecycle
- Access packages and entitlement modelling
- Access reviews and certification campaigns
- Governance workflows and audit reporting

PRACTICAL Run onboarding and an access review.

EVIDENCE JML flow + review report + audit evidence

IDENTITY ENGINEERING EVIDENCE

HYBRID DESIGN • CONDITIONAL ACCESS MATRIX • OIDC/SAML INTEGRATION • CLAIMS MAP • JML FLOW • ACCESS REVIEW

AUTOMATE. CONTROL. ASSURE.

The final month extends identity engineering into application onboarding, lifecycle automation, privileged access, consulting practice, cloud controls and governance evidence.

06 ENTERPRISE IDENTITY DELIVERY

ONBOARD • AUTOMATE • CONTROL PRIVILEGE • ADVISE • ASSURE

I04 Application Onboarding & Enterprise SSO

- Discovery, design and connector selection
- SAML/OIDC onboarding and entitlement tests
- Provisioning, validation and BAU handover

BUILD Onboard and test an enterprise app.

I05 IAM Automation & Integration Engineering

- Microsoft Graph, PowerShell and REST APIs
- SCIM, lifecycle automation and reconciliation
- Exception reporting and operational runbooks

BUILD Automate provisioning and reconciliation.

C Cloud Security

- Azure/AWS IAM, RBAC and least privilege
- Networking, monitoring, keys and secrets
- Containers, Kubernetes and Terraform

TOOLS: Defender • GuardDuty • CloudTrail

I06 Privileged Access Management

- JIT access, vaults and privileged lifecycle
- Session monitoring and secrets management
- Break-glass, cloud and DevOps integration

BUILD Design PAM controls and JIT access.

I07 IAM Consulting & Operations

- Discovery workshops and requirements
- Architecture, standards and support models
- Change control and customer communication

BUILD Present a solution and support runbook.

G Governance, Risk & Compliance

- Risk analysis and treatment
- NIST CSF and ISO 27001 control mapping
- Audit evidence, policies and reporting

TOOLS: Purview • ServiceNow GRC • Archer

MONTH 6 CAPSTONE

BUILD AN ENTERPRISE IDENTITY PLATFORM

SSO • GOVERNANCE • PROVISIONING • PAM • AUDIT EVIDENCE • ARCHITECTURE DEFENCE

PRACTICE BECOMES PROOF.

Every stage combines guided labs, realistic scenarios and documented evidence so students finish with a portfolio they can explain, demonstrate and defend.

01 **BUILD** guided lab → 02 **ANALYSE** evidence → 03 **AUTOMATE** repeatable task → 04 **DOCUMENT** technical record → 05 **PRESENT** career story

CORE LAB ENVIRONMENTS

01 Infrastructure Lab

- Windows Server, Ubuntu and Active Directory
- VirtualBox/VMware and Wireshark
- Network, firewall and directory configuration

OUTPUT Lab architecture, builds and diagrams

02 Security Operations Lab

- Sentinel, Defender XDR, Splunk or ELK
- KQL, dashboards and detection rules
- Phishing, malware and threat hunting cases

OUTPUT Detections, incidents and hunt reports

03 Identity Engineering Lab

- Entra ID, Microsoft 365 and Keycloak
- OIDC, SAML, SCIM and Conditional Access
- Optional IGA and PAM demonstrations

OUTPUT Integration packs and governance evidence

04 Automation & Dev Tools

- Python, PowerShell and Microsoft Graph
- REST APIs, JSON, YAML and Postman/Bruno
- GitHub or institute-managed repository

OUTPUT Scripts, API collections and runbooks

EMPLOYER-READY EVIDENCE

YOUR DIGITAL PORTFOLIO

01 **Infrastructure** — Builds, diagrams and packet analysis
02 **Security Operations** — Dashboards, detections and investigations
03 **Identity & Access** — SSO, governance, automation and PAM
04 **Governance** — Risk, controls, audit evidence and policies
05 **Professional** — GitHub, write-ups, CV and presentation

UPDATE THE PORTFOLIO THROUGHOUT ALL SIX MONTHS

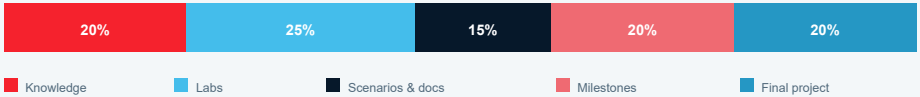
RESPONSIBLE LAB PRACTICE

Test only in authorised environments • use fictional or sanitised data • never publish passwords, tokens or personal information

SHOW WHAT YOU CAN DO.

Progress is measured through knowledge, practical execution, technical evidence and the ability to explain decisions - not by attendance alone.

RECOMMENDED ASSESSMENT WEIGHTING



THREE MILESTONE DEMONSTRATIONS

M2 Small Company Cyber Lab

Infrastructure, network, AD, packet analysis and scripts.

M4 Identity Platform or Mini SOC

SSO, MFA, monitoring, investigation and runbook evidence.

M6 Specialisation Capstone

SOC or IAM engineering pack, portfolio and presentation.

PROGRESSION REQUIREMENTS

- Meet the institute pass mark in knowledge and practical components.
- Complete every milestone project using your own verifiable evidence.
- Finish the specialisation project, portfolio and final presentation.

A Admissions Guidance

- Designed for motivated learners pursuing practical cyber security capability.
- Basic digital confidence and commitment to an accelerated six-month workload are recommended.
- Formal entry checks, documents, schedules and intake details are confirmed by Admissions.

S Learning Support

- Instructor-led explanations and demonstrations.
- Supervised laboratories and scenario feedback.
- Semester reviews, portfolio coaching and presentation practice.
- Academic and technical concerns can be raised with the programme team.

ACADEMIC INTEGRITY

AI tools and references may be used where permitted, but every submission must be verified, understood and defended by the student.

LEARN FROM INDUSTRY PRACTITIONERS.

The lecturer panel connects technical principles with current enterprise identity, cyber security and consulting practice through demonstrations, case studies and guided labs.



Anas Ilyas

Associate Lead - IAM & Cyber Security
Sysco LABS Sri Lanka

An identity and cyber security practitioner with experience across enterprise access, identity operations and professional education. He holds an MSc in Cyber Security and industry learning across Cisco, Azure and IT service management.

CORE EXPERTISE

- IAM operations and access management
- Active Directory and Microsoft Entra ID
- Cloud and enterprise security
- Career-focused technical practice

TEACHING FOCUS
IDENTITY & CYBER SECURITY

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Gayan Gunawardana

Identity & Access Management Consultant
OneIAM | CISSP-certified

An IAM consultant with more than a decade of experience designing and delivering enterprise identity solutions. His work spans modern authentication, federation, Microsoft Entra ID, WSO2 and identity architecture.

CORE EXPERTISE

- Identity architecture and strategy
- OAuth 2.0, OIDC and SAML
- Federation, SSO and passwordless access
- IAM consulting and solution delivery

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INDUSTRY-LED LEARNING

Enterprise scenarios • live demonstrations • technical feedback • portfolio guidance • career perspective